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The effectiveness of VR-based human anatomy simulation training for undergraduate medical students

Sana Ben Hammouda^{1*}, Maher Maoua¹ and Majed Bouchahma²

Abstract

This study evaluated the effectiveness of a virtual reality (VR) human anatomy simulation training program for undergraduate medical students in Tunisia. Pre- and post-training questionnaires assessed student perceptions of the platform's accuracy, realism, ease of navigation, engagement, support for memorization, and its impact on reducing exam-related anxiety. A total of 179 students from the Faculty of Medicine of Sousse participated in the study. Findings indicate that VR anatomy simulation significantly enhances learning outcomes and is highly recommended as a supplementary tool alongside traditional instructional method. Notably, the platform is open-source, cost-effective, and globally scalable—designed to be easily adapted across diverse educational settings. By aligning with international efforts to expand equitable access to advanced learning technologies, this study contributes meaningful insights for both resource-limited and high-resource institutions seeking innovative approaches to anatomy education.

Keywords Metaverse, Human anatomy, Medical education, Immersive education

Introduction

Computer systems have dramatically advanced various fields [1–3] as well as medical diagnostic accuracy and treatment efficacy through the development and implementation of Medical Decision Support Systems (MDSS). These systems leverage established clinical practice guidelines and vast datasets to assist clinicians in making informed decisions, ultimately leading to improved patient care. MDSS are designed to reduce errors, optimize treatment plans, and enhance overall patient outcomes by providing clinicians with evidence-based recommendations and alerts [4–6]. A key innovation

in medical education that complements the benefits of MDSS is immersive learning, which leverages virtual reality (VR) technology to offer students simulated or real-world experiences. The medical field, with its complex and critical nature, stands to benefit greatly from immersive learning [7, 8]. Several studies have demonstrated the effectiveness of immersive learning in medical education. For example, a study conducted by Bork et al. (2009) found that VR simulators can significantly improve the technical skills of medical students in minimally invasive surgery [9]. Another study by Maresky et al. (2018) found that immersive learning can improve the confidence and decision-making skills of anesthesiology residents [10].

Human anatomy is a fundamental component of medical education. Traditionally, cadaver dissection has been the primary method for studying anatomical structures, enabling students to explore internal organs and tissues

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through touch and vision for a deeper understanding of the human body [11–13]. However, access to cadavers is often limited due to high costs, scarcity of donated bodies, and the time-consuming processes involved in embalming, dissection, and disposal [14]. Ethical considerations also play a role, particularly in the handling and disposal of human remains in dissection rooms [15].

As a result, many institutions rely on artificial body parts for learning, which can be costly, particularly in low-income countries where labs can only accommodate a limited number of students per scheduled class, thus restricting access. While students often use textbooks and videos as supplements, they still face challenges in mastering this critical subject [16].

To address these limitations, the use of the 3D Anatomage Table has gained widespread popularity. A study conducted at the Faculty of Medicine, Umm Al-Qura University in Makkah, Saudi Arabia, surveyed 78 students and highlighted several advantages of the Anatomage Table over traditional cadaver dissection. This tool offers highly accurate three-dimensional views of human body parts, allows for longitudinal, sagittal, and horizontal sections, and integrates CT/MRI imaging, all of which contribute to a deeper understanding of anatomy. Additionally, it provides unlimited use without the need for time-consuming cadaver preparation and disposal [17].

However, the Anatomage Table lacks tactile experience and cannot display anatomical variations such as supernumerary muscles and different nerve patterns [17, 18]. Moreover, its use is limited by the number of students it can accommodate at a time, and it requires students to be physically present at the university, which may restrict flexible learning opportunities [19, 20].

Recent advancements in VR-based medical education have demonstrated the potential of immersive technologies to address the limitations of traditional methods. For instance, M Kadri et al. [21] developed an Immersive Virtual Anatomy Laboratory (IVAL) that combines VR with serious games, resulting in high user satisfaction and improved understanding of skeletal anatomy. Their comparative study further revealed that VR significantly enhances learning outcomes, including reduced learning time and higher exam scores, compared to traditional methods [22]. Additionally, A Neyem et al. [23] showed that integrating VR into traditional anatomy labs increases student engagement and improves their ability to understand complex anatomical structures and spatial relationships, offering a repeatable and ethical alternative to cadaver dissection. Finally, M Kadri et al. [24] conducted a longitudinal study on collaborative VR-based education, demonstrating that VR not only enhances long-term knowledge retention but also fosters collaboration among students, highlighting the importance of collaborative learning theories in immersive education.

Together, these studies provide strong evidence for the effectiveness of VR as a supplementary tool in medical education, offering a more engaging, flexible, and impactful learning experience.

Despite these advancements, a significant gap remains in the availability of affordable and adaptable VR-based anatomy learning platforms that cater to diverse student needs. Existing tools are often costly, limited in accessibility, or lack customization to fit specific curriculum requirements. Furthermore, many platforms focus primarily on visualization rather than incorporating interactive assessments and adaptive learning strategies.

This study aims to develop a novel open source metaverse platform for Human anatomy education that provides an interactive, flexible, and cost-effective solution for medical students. Unlike existing VR tools, this platform will integrate personalized learning pathways, real-time feedback, and collaborative features, ensuring a more comprehensive and engaging educational experience. The platform will also incorporate an advanced virtual examination system that allows students to test their knowledge through randomized quizzes, interactive case studies, and real-time assessments. This system will not only enhance knowledge retention but also prepare students for real-world medical evaluations by simulating exam conditions in an immersive and interactive environment. By providing an adaptable and scalable solution, this metaverse platform will bridge the gap between traditional anatomy education and modern technological advancements, ensuring a well-rounded learning experience.

The proposed metaverse platform aims to revolutionize anatomy education by offering:

- **Enhanced Interactivity vs. Passive Learning:** Unlike traditional anatomy education, which relies heavily on cadaver dissection and textbooks, the metaverse platform allows students to manipulate 3D anatomical structures, engage in virtual dissections, and simulate medical procedures, making learning more hands-on and engaging [25–27].
- **Risk-Free Practice vs. Resource Constraints:** Traditional methods require physical cadavers, which are costly, scarce, and raise ethical concerns. The metaverse platform eliminates these issues by offering unlimited virtual dissections and practice sessions without material constraints, reducing dependency on cadavers and ensuring repeated exposure to complex anatomical structures [28, 29].
- **Accessibility and Flexibility vs. Fixed Learning Environments:** Traditional anatomy labs require students to be physically present at specific locations and times. The metaverse platform provides anytime, anywhere access, allowing students to integrate

learning into their schedules and revisit content as needed, enhancing knowledge retention and enabling self-paced study [30].

- **Comprehensive Assessment vs. Limited Feedback:** Traditional assessments often rely on written exams and practical tests with limited interactivity. The metaverse platform integrates real-time assessments, interactive case studies, and AI-driven feedback mechanisms, ensuring continuous evaluation and targeted improvement suggestions.

Through these advancements, the study objectives are to:

- Develop an open source metaverse platform tailored for undergraduate medical students to enhance anatomy learning.
- Integrate interactive 3D anatomical models that allow students to explore complex structures with high precision.
- Incorporate a virtual examination system with randomized quizzes to assess knowledge retention and exam preparedness.
- Ensure accessibility by offering the platform free of charge, enabling equal opportunities for students worldwide.

By addressing these key areas, this study seeks to provide a more interactive, accessible, and effective supplement to traditional anatomy training.

Systematic reviews and meta-analyses

A comprehensive search of relevant literature was conducted using databases such as PubMed, Web of Science, Scopus, and the Cochrane Central Register of Controlled Trials (CENTRAL). Search terms included “virtual reality,” “anatomy education,” “medical students,” “simulation training,” “augmented reality,” and “mixed reality.” The search focused on articles published between January 2023 and November 2024 to ensure the inclusion of the most recent evidence. Studies included systematic reviews, meta-analyses, randomized controlled trials (RCTs), and other experimental studies evaluating the effectiveness of VR and AR in anatomy education. The review process adhered to PRISMA guidelines.

Several recent systematic reviews and meta-analyses have investigated the effectiveness of VR and Augmented Reality (AR) in anatomy education. Paloma Garca-Robles et al. [31] conducted a meta-analysis of 27 experimental studies involving 2199 health sciences students and found that XR technologies (VR and AR) yielded higher knowledge gains than traditional approaches, especially when used as supplemental learning resources. Their analysis also revealed that 80% of students perceived XR devices as useful for learning anatomy. Sajjad Salimi et al. [32]

performed a meta-analysis of 24 randomized controlled trials and concluded that VR had a moderate and significant effect on improving knowledge scores compared to other methods. However, they also noted a high degree of heterogeneity across the included studies, suggesting the need for further research to identify variables impacting the efficacy of VR and AR in anatomy education. Ally Williams et al. [33], Williams conducted a systematic review comparing AR with other teaching methods in learning anatomy. While they acknowledged AR's potential, they found that definitive outcomes from the current literature are limited by the heterogeneous nature of the studies and inconsistent use of terminology.

Jonathan Awori et al. [34] compared the effectiveness of VR and 3D-printed models in teaching cardiac anatomy to residents and nurse practitioners. The study found that trainees endorsed VR as more effective than 3D-printed models and traditional instruction for enhancing their understanding of cardiac anatomy and associated pathophysiology. Subin Park et al. [35] conducted a systematic review and meta-analysis of immersive technology-based education for undergraduate nursing students, finding that VR-based education was more effective than traditional education in knowledge attainment, confidence, and self-efficacy. Areej Banjar et al. [36] reviewed experimental studies on the effectiveness of mixed reality in higher education and found that mixed reality technologies have the potential to enhance learning experiences, particularly in medical and health sciences.

In summary, recent systematic reviews and meta-analyses highlight the growing evidence supporting VR and AR in anatomy education, demonstrating improved knowledge retention and positive student perceptions. However, variability in study designs and terminology suggests the need for further research to refine best practices. These findings informed us of our study design and methodology, which we detail in the following section.

Methodology

This study aimed to evaluate the effectiveness of a VR-based human anatomy simulation training program for undergraduate medical students in Tunisia. The primary objective was to assess the impact of VR-based learning on students' understanding of anatomical structures and to explore how it compares to traditional learning methods.

Study design and sample selection

To determine the appropriate sample size (S), Cochran's equation was applied, ensuring statistical validity with a 95% confidence level and an expected proportion (P) of 0.5. This calculation was chosen because the total number of students in the target population was known, relatively small, and limited, making it a suitable method

for accurately estimating the required sample size while maintaining representativeness. The calculation was based on recent statistics from the Ministry of Higher Education, which detail the number of undergraduate students over the past two years. By using this approach, we ensured that the selected sample accurately reflected the broader student population, allowing for reliable conclusions about the effectiveness of VR-based anatomy training.

$$S = \frac{X^2 \cdot N \cdot P \cdot (1 - P)}{d^2 \cdot (N - 1) + X^2 \cdot P \cdot (1 - P)}$$

X = 1,96 (1% error, 95% standard confidence).

P: Population portion (30-50%).

P = 50%.

d = 0.05 (standard degree of accuracy).

N = Number of estimated populations.

To ensure a fair and unbiased selection process, a simple random sampling (SRS) method was used to select 179 students from Ibn Al-Jazzar Faculty of medicine, Sousse. This approach provided each student with an equal chance of selection, minimizing bias and ensuring a representative sample. Students were then randomly assigned to either the experimental group (VR-based training) or the control group (traditional learning methods) using a computer-generated randomization list. This process ensured an even distribution of participants across both groups, reducing selection bias and allowing for a balanced comparison of outcomes.

We used the following python code to create the two groups. The script will ensure an even distribution and output the assignments in a structured format (Fig. 1).

Eligibility criteria

The study focused on first- and second-year medical students, as the VR scenarios were specifically designed to align with their theoretical knowledge and curriculum. This selection ensured that the anatomical content was relevant and appropriate for their level of study. Eligible participants met the following inclusion criteria:

- Enrollment in the first or second year of medical school to ensure alignment with the anatomy curriculum.
- No prior experience with VR-based anatomy training to avoid bias in learning outcomes.
- No medical conditions that could interfere with VR use, such as severe motion sickness, epilepsy, or visual impairments that could compromise the immersive experience.
- Willingness to participate and provide informed consent, ensuring ethical compliance.

Setting and recruitment

The study was conducted within the medical school's anatomy department, where students regularly engage in both theoretical and practical anatomy courses. Recruitment was carried out through student emails and in-class announcements. Interested students were invited to an informational session where they were briefed on the study's purpose, procedures, potential benefits, and risks.

Those who met the eligibility criteria and agreed to participate provided written informed consent.

Demographics and gender distribution

Participants' ages ranged from 18 to 23 years, with a mean age of 20.3 years and a standard deviation of 1.4 years. To ensure gender representation, the closest possible numbers of male and female students were selected, resulting in a final sample of 98 females and 81 males, closely reflecting the university's enrollment distribution. This balance helped ensure that findings were generalizable across both genders.

Normality testing & effect size reporting

In our study, we conducted normality testing to assess the distribution of the data before performing statistical analyses. The Shapiro-Wilk test was used to determine whether the data followed a normal distribution. This test was applied to key variables such as training session attendance and time spent in the VR environment, ensuring that the data met the assumptions for parametric tests and validating the strength of our results.

To quantify the magnitude of differences between VR-based training and traditional learning methods, we reported effect sizes. For comparisons using t-tests, we applied Cohen's d, which provides a standardized measure of the difference between the two groups. By incorporating both normality testing and effect size reporting, we ensure a more comprehensive and transparent interpretation of the data, allowing for a clearer understanding of the effectiveness of VR-based training compared to traditional methods.

Handling of missing data

In cases where data were missing, we applied listwise deletion, ensuring that only complete datasets were analyzed. This approach helps maintain the integrity and validity of the results while minimizing potential biases that could arise from incomplete data.

Confidence intervals and p-values

We have ensured that confidence intervals (95% CI) and p-values are consistently reported throughout the results section. This provides a clearer interpretation of statistical significance and the precision of our estimates, particularly when comparing the outcomes of VR-based

```
import pandas as pd
import random
# Load participants from an Excel file
file_path = "participants.xlsx"
sheet_name = "Sheet1"

# Read the participant names from the first column of the Excel
file
df_participants=pd.read_excel(file_path,sheet_name=sheet_n
ame)
participants = df_participants.iloc[:, 0].dropna().tolist()

# Shuffle the list to ensure randomness
random.shuffle(participants)

# Split into two groups
midpoint = len(participants) // 2
experimental_group = participants[:midpoint]
control_group = participants[midpoint:]

# Create a DataFrame for better visualization
df_randomized = pd.DataFrame({
    "Participant": participants,
    "Group": ["Experimental"] * midpoint + ["Control"] *
(len(participants) - midpoint)})

# Save the assignments to an Excel file
df_randomized.to_excel("randomized_groups.xlsx",index=False)
```

Fig. 1 The script to randomly generate experimental and control groups

training and traditional learning methods. By consistently including these values, we aim to enhance transparency and facilitate a more comprehensive understanding of the effectiveness of the interventions.

Confounding variables

To control confounding variables (e.g., prior human anatomy knowledge, gender distribution...), we conducted multivariate regression. This analysis ensured the observed effects were attributed specifically to the VR intervention rather than external influences.

Questionnaire development and validation

The questionnaire used in this study was designed to assess students' perceptions of the VR simulation and compare it to traditional learning methods. The eight questions focused on key aspects such as engagement, memorization, and anxiety reduction.

To ensure validity and reliability, a professional team of teachers and faculty members with expertise in medical education and human anatomy reviewed and validated the questionnaire. Their feedback was incorporated to refine the questions, ensuring they accurately captured

students' experiences with the VR simulation and its potential advantages over traditional methods.

The questionnaire was divided into two sections.

The first section comprised Likert scale questions (from 1 to 5) to quantitatively assess the effectiveness of the VR training. Each question targeted specific factors, including:

Engagement: How engaging did you find VR training?

(1 = Not engaging, 5 = Highly engaging)

Effectiveness: How effective was VR training in helping you learn anatomy?

(1 = Not effective, 5 = Extremely effective)

Anxiety Reduction & Memorization: Do you think virtual training could alleviate your apprehension and improve memorization of human anatomy?

(1 = Not helpful, 5 = Very helpful)

The second section consisted of open-ended questions, allowing students to share their perspectives, highlight strengths, and provide constructive feedback on improving the VR platform.

Pre- and post-simulation responses were analyzed to identify shifts in perception and assess the impact of VR training on students' learning experiences.

Study procedure

Pre-Simulation phase

Students attended a 15-minute lecture on using the VR application to familiarize themselves with the immersive experience. Afterward, they completed a pre-simulation survey to assess their initial perceptions of VR-based learning, including memorization, pre-exam anxiety.

VR simulation and assessment

Before the study, a preliminary test was conducted to determine the optimal duration for each session, ensuring that students had sufficient time to engage in the virtual environment without experiencing cognitive overload or fatigue. Based on the results, it was decided that each student would have 45 min to explore, learn, and complete the quiz. This timeframe allowed for meaningful interaction with the VR platform while maintaining focus and engagement.

The selected students explored the virtual human anatomy lab, with a primary focus on the skeletal and muscular systems. Inside the VR environment, they had access to highly detailed 3D anatomical models, which they could interact with using various tools.

Students were able to rotate, zoom in, and manipulate structures to examine internal relationships from multiple perspectives, simulating a real dissection experience. This hands-on approach provided an immersive learning experience, reinforcing theoretical knowledge through active exploration and visualization.

Following their interaction with the anatomical models, students were required to complete a six-question quiz integrated into the VR platform. Each question was designed to assess their comprehension of key anatomical structures and concepts. To maintain the integrity of the assessment, each question could only be answered once, encouraging students to apply critical thinking and recall their observations accurately. These quizzes served as realistic simulations of formal assessments, helping students familiarize themselves with exam-style questioning in an interactive setting.

Upon completing the quiz, students received immediate feedback on their performance. This feedback highlighted correct and incorrect responses, allowing students to review their mistakes, reinforce learning, and address any gaps in their understanding. Such an approach not only enhanced knowledge retention but also provided a self-paced learning experience that encouraged continuous improvement.

At the end of the simulation, students exited the virtual lab by selecting the "Go Home" button, which redirected them to the main menu, officially concluding their session (Fig. 2).

This structured workflow ensured that students followed a guided yet flexible learning path, maximizing their engagement with the VR platform while facilitating a seamless transition between exploration, assessment, and review (Fig. 2).

Instructors may play a crucial role in the learning process, adapting their teaching styles to integrate VR into anatomy education. The VR session was displayed on a smart TV or other screens, allowing instructors to monitor students' progress, identify learning gaps, and provide real-time assistance.

This setup enabled a more interactive teaching experience, where instructors could guide students, clarify misunderstandings, and reinforce complex anatomical concepts. Additionally, the multiplayer option introduced a new dimension to teaching by allowing instructors to join the virtual environment as avatars. In this setting, they could directly interact with students, provide instant feedback, and lead discussions in a dynamic and engaging virtual classroom. This feature transformed traditional teaching into an interactive and collaborative learning experience, enhancing student engagement and comprehension through a more immersive and personalized approach.

Post-simulation phase

Students completed a post-simulation survey, which included the same eight questions as the pre-survey, assessing various aspects of the VR training, such as accuracy, realism, ease of navigation, engagement, impact on memorization, and reduction of exam-related

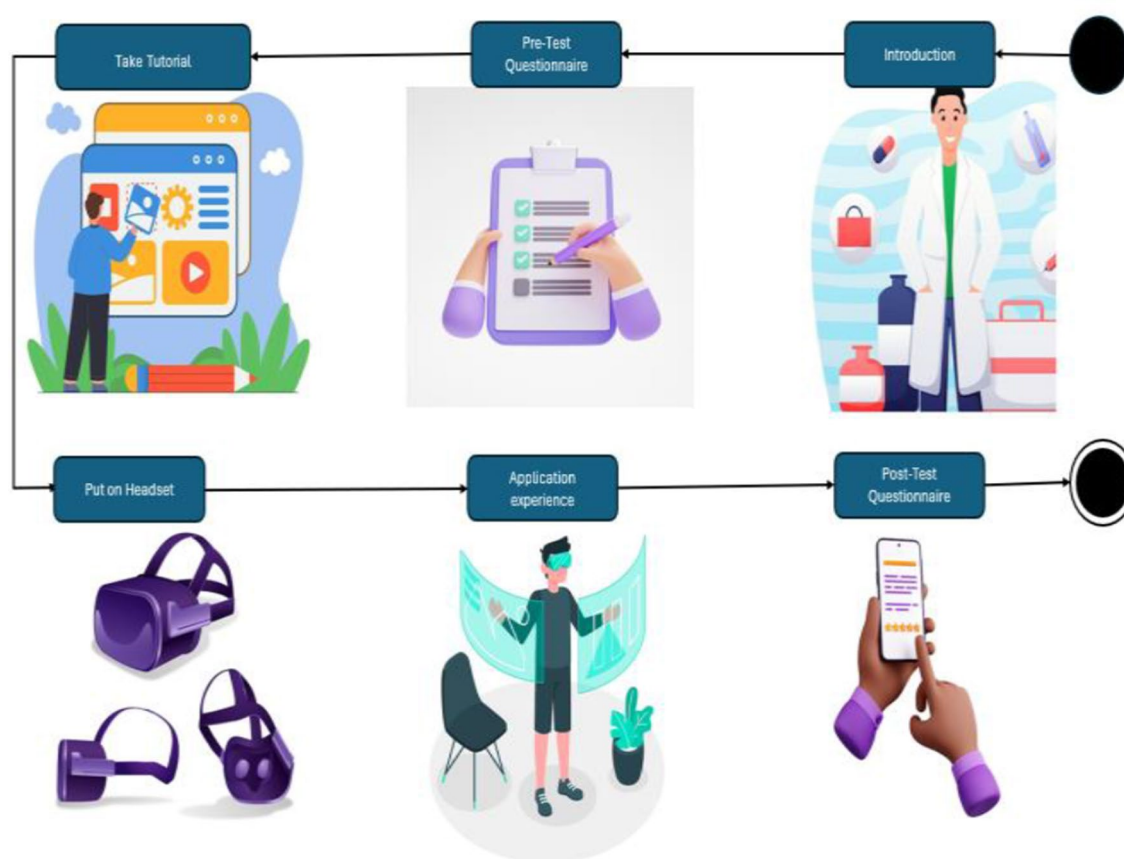


Fig. 2 Overall process of the study

anxiety. An open-ended section allowed students to share feedback on the VR platform and its influence on their anatomy learning experience, helping guide future improvements. Most students remained motivated to continue using the platform throughout the academic year. To further strengthen our study, we maintained contact with all participants for follow-up at six months and one year. This follow-up also allowed us to evaluate long-term knowledge retention, providing valuable insights into the lasting impact of VR-based learning on students' understanding of human anatomy.

Data analysis

Survey responses were collected electronically over the course of one academic year and analyzed using Microsoft Excel. The data was analyzed through frequencies and percentages to evaluate the impact of the VR simulation and its perceived benefits on students' learning experiences.

A comparative analysis was conducted between the pre- and post-survey results to assess changes in student perceptions, particularly regarding the effectiveness of VR in comparison to traditional learning methods.

Our statistical models and findings were validated by an independent biostatistics expert, ensuring robustness and accuracy of the analytical framework.

Before participating in the study, all students signed a written informed consent form after being provided with a detailed explanation of the study's objectives, procedures, potential benefits, and any associated risks. An informational session was conducted to ensure that participants fully understood the research process, their voluntary participation, and their right to withdraw at any time without consequences.

The study strictly adhered to ethical guidelines, including the Declaration of Helsinki and institutional research ethics standards. Ethical approval was obtained from the university's ethics committee (DECISION N°256 [Ref: CEFMS 256/2024]), ensuring compliance with all relevant regulations. Confidentiality and anonymity of participant data were maintained throughout the study, with all collected information securely stored and used solely for research purposes.

By implementing these measures, we ensured that participants' rights, safety, and privacy were fully protected throughout the research process.



Fig. 3 Application home screen

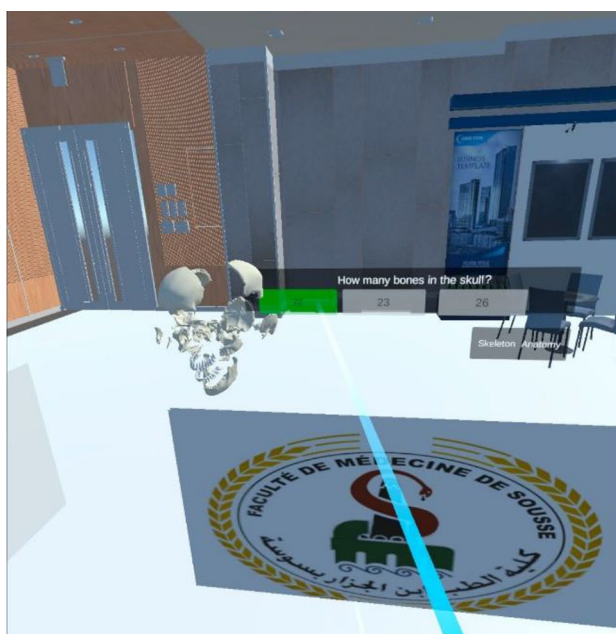


Fig. 4 Application screenshot on Meta quest 2 showing an assessment question

Software development

The metaverse VR simulator software was developed by authors using, Unity3D. All 3D objects in the software were developed in Blender or downloaded from Sketch fab (www.sketchfab.com) whereas programming of simulator software was programmed on Unity3D. Agile software development technique [37] was used to build the simulation software. Agile software development is the idea of iterative development where requirements and

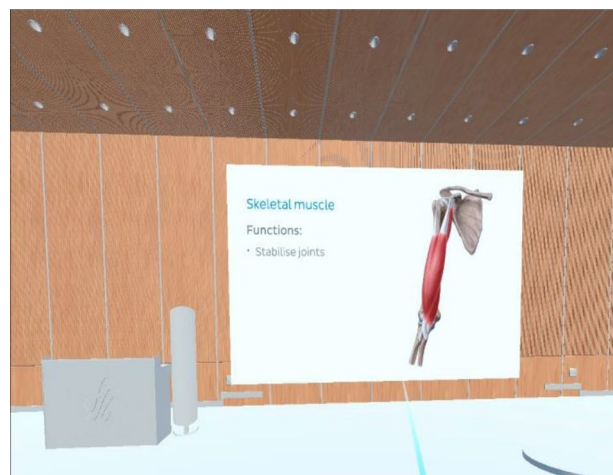


Fig. 5 Application screenshot on Meta quest 2 showing a video content on human anatomy

solutions evolve through collaboration between self-organizing and cross-functional teams.

Users will navigate through various labs, each represented by different Unity scenes. This article focuses specifically on muscle and skeletal anatomy, as these are fundamental, straightforward, and essential topics for undergraduate students.

The home screen of the application is a hall with doors to access the different labs. Students will select one of the labs using their pointing devices (Joysticks) (Fig. 3).

Students will be assessed with 6 quiz questions, each of which can only be answered once. Once a question is answered, it will be disabled. If the student selects the correct answer, it will be highlighted in green, and if the wrong answer is chosen, it will be marked in red (Fig. 4).

In the VR application, students can also enhance their learning by watching instructional videos that cover the anatomy of human muscles and bones. These videos serve as a revision tool, helping students prepare quizzes and reinforcing their understanding of the course material.

This integration of visual content with interactive 3D models offers a comprehensive immersive learning experience (Fig. 5).

Students can use VR controllers or hand tracking to grab, rotate, and move anatomical models. This allows them to closely inspect different parts of the body from various angles (Fig. 6).

At the end of each scenarios students they can see their score before going back to the home screen (Fig. 7).

The technical aspects of the VR platform were carefully evaluated in collaboration with a Unity expert to minimize usability challenges. In case of unexpected issues, exciting and restarting the application can resolve disruptions and ensure a seamless experience.



Fig. 6 Manipulation of the models in the application



Fig. 7 Score screen

Implementing multiplayer functionality within the metaverse platform could further enhance interactivity and improve overall outcomes. However, a persistent limitation remains the platform's dependence on a stable internet connection for optimal performance.

To address this, either maintaining a single-player offline mode or integrating multiplayer with enhanced reconnection protocols could significantly improve the platform's reliability and accessibility.

Discussion and results

The results of the questionnaires demonstrated strong student support for integrating VR anatomy simulations as a supplementary tool in their learning process.

Perceived accuracy and realism

A significant 86% of students found the VR scenarios to be accurate. While accuracy was formally evaluated by an expert team, student feedback remains valuable, as they are the primary users of the technology. Additionally, 72% of students rated the VR scenarios as realistic when compared to the traditional anatomy lab experience, suggesting a strong sense of immersion and authenticity (Fig. 8).

These findings are consistent with international studies highlighting the pedagogical benefits of immersive technologies in anatomy learning [38]. Our results reinforce the idea that VR enhances anatomical comprehension regardless of geographic or institutional context.

Ease of navigation

Navigation within the VR environment was reported as intuitive by most students. Specifically, 39% found it very easy to explore the virtual scenarios, while 31% rated it as easy. Additionally, 27% found it moderately easy. A small percentage of students faced challenges, with 3% reporting difficulty and only 1% finding it very difficult. These results suggest that the VR system is largely user-friendly and accessible. (Fig. 8).

Importantly, our open-access platform was designed to minimize technical barriers, making it deployable across diverse educational institutions with varying technological infrastructures. Unlike many commercial VR platforms that require costly subscriptions or proprietary software, our model is modular, lightweight, and compatible with low-spec VR headsets and PCs. This allows scalability across institutions in both high-income and resource-constrained regions. Moreover, the platform features a multiplayer mode, enabling students to enter virtual sessions alongside their instructors or peers. This collaborative functionality not only supports guided learning and peer-to-peer interaction but also assists students in navigating and utilizing the platform more effectively.

These findings align with international research emphasizing the positive impact of immersive learning on cognitive retention and learner attention. For instance, a study at Qatar University using the 3D-Organon VR platform demonstrated significant improvement in students' anatomical understanding and memorization, with a strong preference for VR over traditional methods [39].

Similarly, research conducted at the University of Saskatchewan in Canada revealed that immersive VR-based neuroanatomy training boosted student motivation and

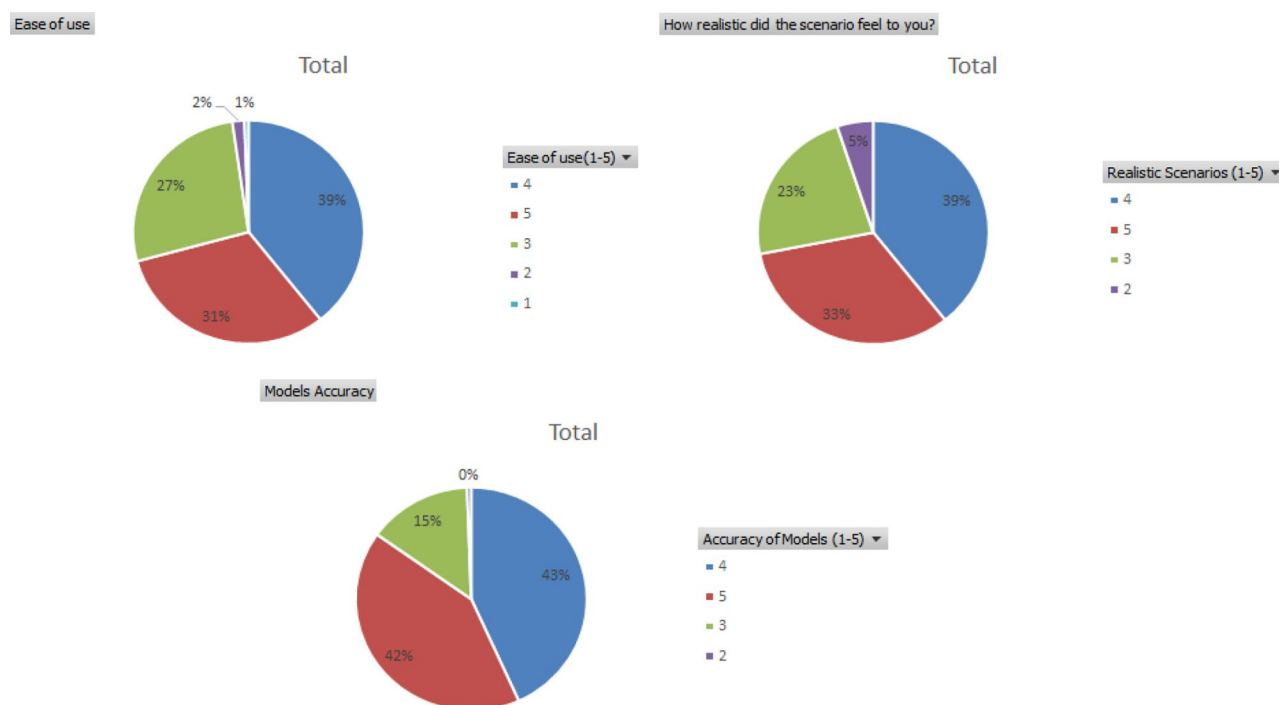


Fig. 8 Students responses for ease of use, how realistic were the scenario and the accuracy of the models after the usage of the application

reduced anxiety, leading to better knowledge retention than paper-based approaches [40]. Furthermore, a meta-analysis of 15 randomized controlled studies confirmed that VR-based anatomy education significantly enhances test scores and long-term retention compared to conventional teaching strategies [41]. Our platform thus complements and extends this global trend, reinforcing the role of immersive VR technologies in transforming medical education by enhancing engagement and learning efficiency across diverse contexts.

Reduce of anxiety

A notable 82% of students believed that VR scenarios could help reduce their anxiety before an anatomy exam. Additionally, immersive learning was seen as a tool that alleviates stress and builds confidence by allowing students to practice in a controlled, interactive environment. This suggests that VR-based training provides a more comfortable, less stressful preparation method for exams (Fig. 9).

By simulating exam conditions in a virtual environment, our platform echoes findings from international literature suggesting that VR can lower cognitive load and improve exam confidence.

The data analysis suggests that VR is perceived as an effective learning tool that enhances memorization and reduces exam anxiety. Students show strong interest in its integration, reinforcing its potential role in medical education (Table 1).

Anxiety reduction

The mean score for “Anxiety before exams” ($M=2.70$, $SD=1.10$) is lower than “Reduction of anxiety” ($M=4.11$, $SD=0.87$), suggesting that VR training might help alleviate students’ pre-exam anxiety.

Interest & integration of VR Students show high interest in using VR for learning ($M=4.39$, $SD=1.02$) and perceive it as an effective tool ($M=4.53$, $SD=0.61$). This indicates strong engagement and potential for curriculum integration.

Memorization & learning efficiency Memorization with Anatomy Model” ($M=4.29$, $SD=0.90$) and “Apprehension and memorization” ($M=4.78$, $SD=0.47$) show strong agreement that VR enhances memory retention and comprehension.

These statistically significant shifts reflect the platform’s potential for broader application. The lightweight nature of the software ensures that this model could be adopted at scale across academic institutions worldwide.

When asked about the overall usefulness of VR as a learning tool, 93% of students (59% strongly agree + 34% agree, as shown in Fig. 10) supported integrating VR scenarios into their curriculum, recognizing its value in enhancing engagement and improving their understanding of complex topics. However, only 51% (29% strongly agree + 22% agree, as shown in Fig. 10) believed that VR simulations could fully replace physical lab sessions.

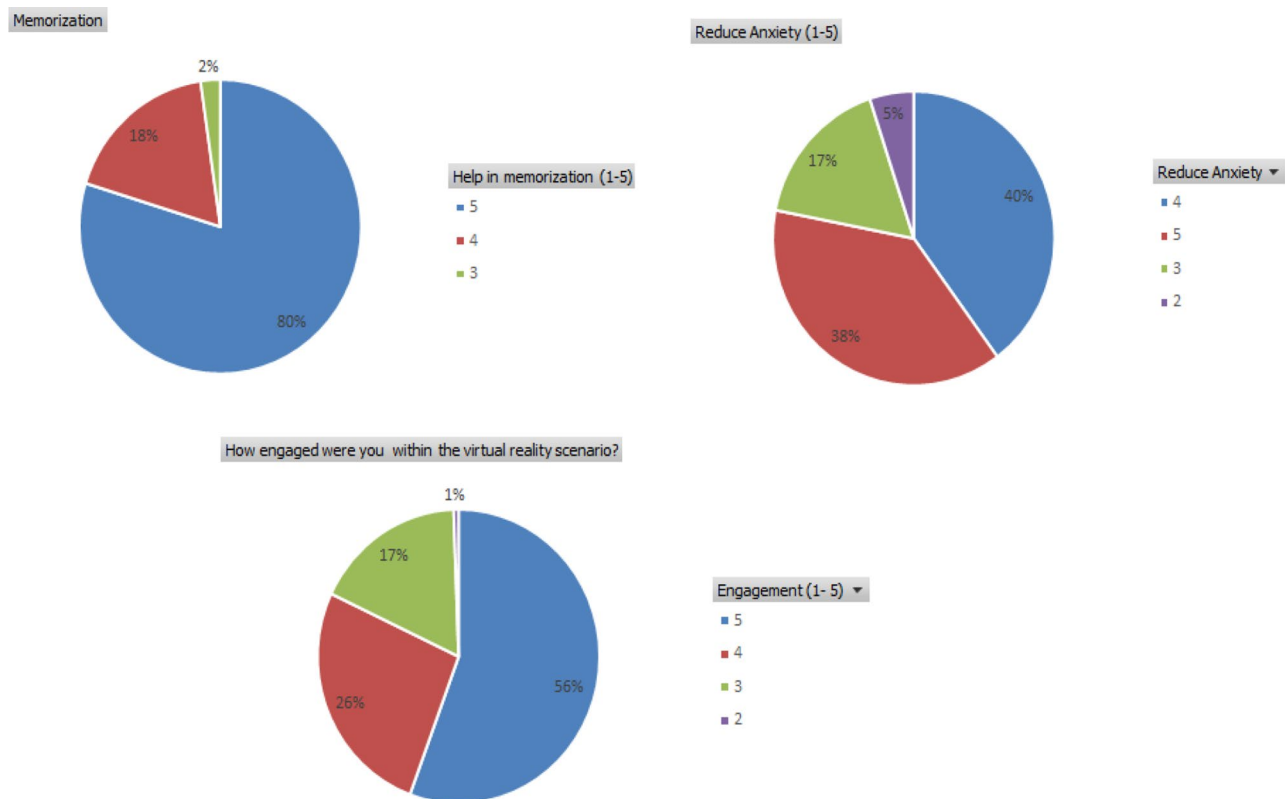


Fig. 9 Students responses for engagement, anxiety and memorization aid after usage of the application

Table 1 Effects of VR on anxiety, interest, and memorization

Factor	Pre-VR Mean (M) ± SD	Post-VR Mean (M) ± SD
Anxiety Reduction	2.70 ± 1.10	4.11 ± 0.87
Interest & Integration of VR	4.39 ± 1.02	4.53 ± 0.61
Memorization & Learning Efficiency	4.29 ± 0.90	4.78 ± 0.47

Additionally, 20% were somewhat supportive of the replacement, while 19% completely disagreed with replacing traditional labs (Fig. 10).

This highlights that, while students appreciate the immersive and interactive nature of VR, they still see hands-on experience in a traditional lab as essential for

skill development and practical application. The findings suggest that students view VR as an effective complementary resource to physical labs, augmenting learning by offering virtual experiences that can reinforce and supplement what they learn through direct interaction with real-world materials.

Many students were enthusiastic about their experience with the VR platform, praising its immersive and engaging nature. Some expressed a desire for a broader range of medical scenarios, including emergency medicine, histology, and more detailed surgery-related content. The overall feedback reflects a high level of satisfaction, with an eagerness for expanded content and increased interactivity.

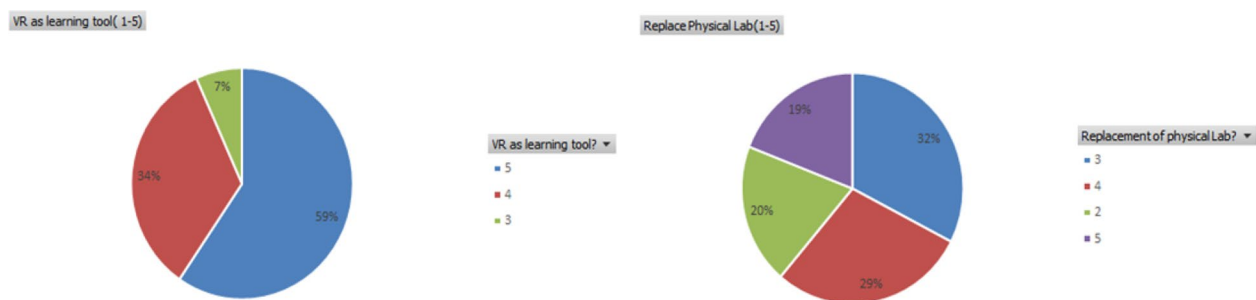


Fig. 10 Students responses for the use VR as additional learning tool or replacement of physical labs

This positive reception shows the platform's effectiveness and its potential to become an even more valuable tool in medical education. Students appreciate how VR enhances their learning, but they see opportunities for it to cover more subjects, making it a comprehensive and versatile resource. Addressing these suggestions would not only enrich the learning experience but also better align the platform with the diverse needs of medical students.

Expanding the content and adding more interactive features can significantly boost the platform's educational value, making it a key asset in training future healthcare professionals. The platform has already made a notable impact, and with these enhancements, it can provide an even more holistic and dynamic learning experience across various medical disciplines, better preparing students for both exams and real-world scenarios.

Individual feedback also reflects the platform's positive impact:

"It was a really amazing immersive experience."

"I felt absolutely engaged, and I didn't even want to exit the extraordinary environment."

"I feel like I'm in a real exam, and I believe this will help boost my knowledge and self-confidence before anatomy assessments."

However, some students, especially those new to VR, shared challenges:

"This is my first time using VR, so I'm still struggling with it."

"It's perfect, but it would be even better if you added more subjects and interactive scenarios, like surgery, emergency cases, and histology."

To summarize, unlike proprietary systems such as 3D Organon or Visible Body, our VR anatomy platform is open-access, making it more accessible and affordable—especially for institutions in low-resource settings. It also features a multiplayer mode, enabling students to interact in real time with instructors or classmates, which fosters collaboration, boosts confidence, and supports guided learning if we compare it with solution like Anatomage Table. Since the platform is delivered entirely in English, it is well-suited for international use and aligns with global education standards.

Conclusion and future work

Our findings support recent research demonstrating the positive impact of immersive VR simulations on medical education. While VR has proven to be a powerful tool for enhancing learning, it is not yet a complete substitute for the hands-on experience of traditional lab work. Still, student feedback from this study shows high satisfaction with the VR platform, emphasizing its strong potential to transform anatomy education. Many students expressed

a desire for more content and interactivity, which we see as an opportunity for growth.

In response, we plan to further develop the platform by integrating new features based on user feedback. One key enhancement will be the addition of an emergency room simulation, allowing students to engage with realistic clinical scenarios in a safe, controlled environment. We also aim to introduce multilingual support to make the platform more accessible to non-English-speaking learners, broadening its reach and impact.

Looking ahead, a cloud-based repository will be established where academic institutions can freely download, customize, and contribute to the library of virtual learning scenarios. This will help foster a global community of educators and learners, united in their efforts to improve medical training through accessible technology.

By significantly improving students' understanding of anatomy, reducing exam-related anxiety, and increasing engagement, VR-based training presents a promising complement to traditional education methods. Thanks to its open-source, scalable design, our platform is particularly well-suited for global use—especially in regions where commercial VR tools remain out of reach. This study offers an inclusive, adaptable model that can be replicated and expanded by institutions worldwide. Future efforts will focus on growing the scenario database, enhancing language support, and building partnerships with medical schools across the globe.

Abbreviations

VR	Virtual reality
3D	Three dimensional
CT/MRI	Computed tomography/ magnetic resonance imaging
MDSS	Medical decision support systems
IVAL	Immersive virtual anatomy laboratory
SRS	Simple random sampling

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12909-025-07402-5>.

Supplementary Material 1

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Author contributions

All authors contributed to the design, implementation, analysis, and writing of this study. Each author reviewed and approved the final manuscript.

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Data availability

The data is available for private use or upon request.

Declarations

Ethics approval and consent to participate

This study received written ethical approval from the Ethics Committee of the Faculty of Medicine in Tunisia. Written consent was obtained from all participants prior to their involvement in the research study.

Consent for publication

The "Consent for Publication" section is not applicable to this manuscript. Informed consent for participation was obtained from all participants, and their privacy is guaranteed as all students are anonymized.

Competing interests

The authors declare no competing interests.

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